Pixel-Based Optical Navigation in Cislunar Space

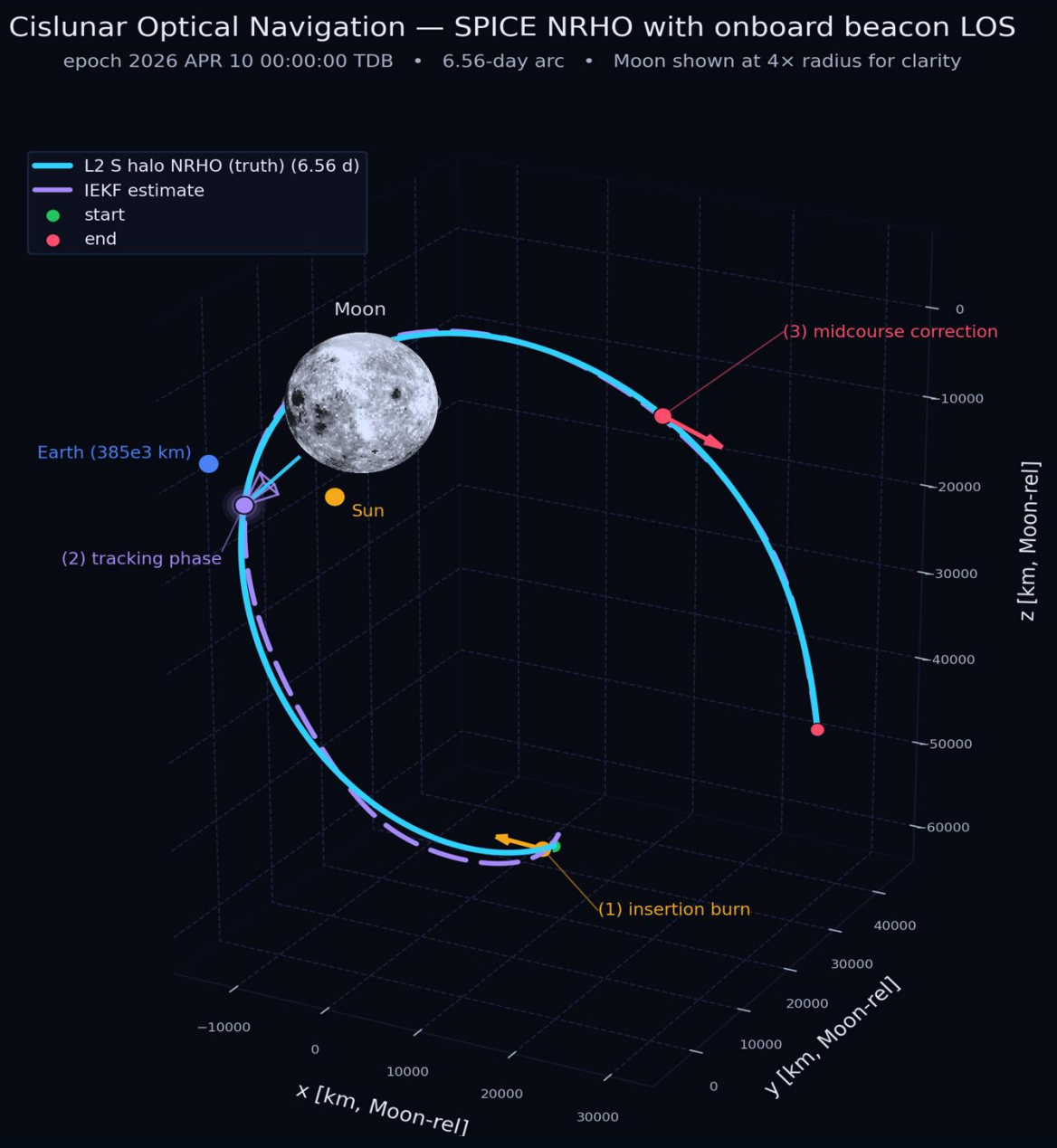
Bearing-Only State Estimation

Active Tracking

Midcourse Correction

Enabling autonomous navigation beyond Earth orbit using onboard vision

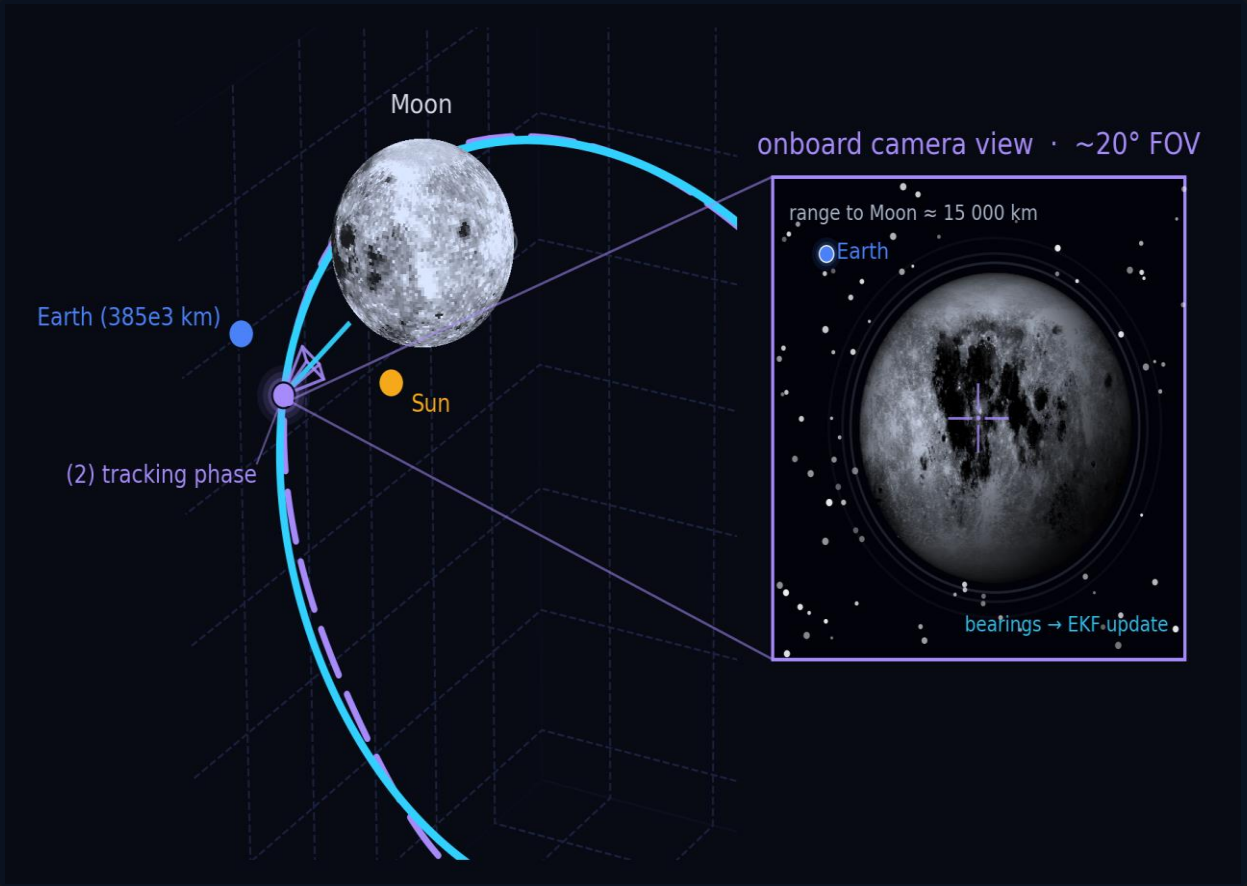
Phases 00-07 Demo for ASTE 581 Final



V. Krishnakumar (GNC/CV Engineer)

Why use a Camera for Cislunar Navigation?

In cislunar space, navigation cannot rely on strong, continuous GPS-like positioning support, so onboard vision becomes an attractive source of autonomous navigation information



Navigation Challenge

- Spacecraft may not have continuous access to strong GPS-like navigation support
- Ground-based tracking remains valuable, but onboard autonomy becomes increasingly important

Why a camera?

- Passive, lightweight, and power-efficient
- Provides continuous geometric information whenever the target is visible
- Naturally supports landmark tracking and horizon sensing

Tradeoff

- Camera does not directly measure range (ρ)
- Provides image-plane info (u,v) that becomes a line-of-sight direction
- Missing direction must be recovered from dynamics and repeated observations

Project idea: use onboard vision to replace or supplement weak navigation observables in cislunar space

Cislunar Optical Navigation — Software Pipeline

SPICE-backed IEKF with autonomous camera pointing (Phase 7)

This talk focuses on Phases 02–07 · Phase 08 closes as a future-facing extension

Truth · Measurements · Estimation · Guidance · Control

ZONE 1 · Dynamics / Truth

00–01

Ephemeris / Scenario
JPL DE442 kernels

Dynamics Propagation
Integrated state + STM

ZONE 2 · Vision / Measurement generation

04–05

Camera Model
Pinhole + FOV projection

Target Detection
Centroid in image plane

Pixel → Bearing
(u, v) → unit LOS

03

Bearing Measurement
Residual + covariance

ZONE 3 · Estimation

07

Phase 7 · Camera Pointing Logic
Repoints camera from filter estimate
so the target stays inside the FOV

CLOSED-LOOP NAVIGATION

ZONE 4 · Guidance / Correction

02

Apply Correction + Repropagate
Δv injected · truth updated

Midcourse Targeting Solve
Δv to reduce terminal miss

03

State Estimate
$\hat{x}$, P

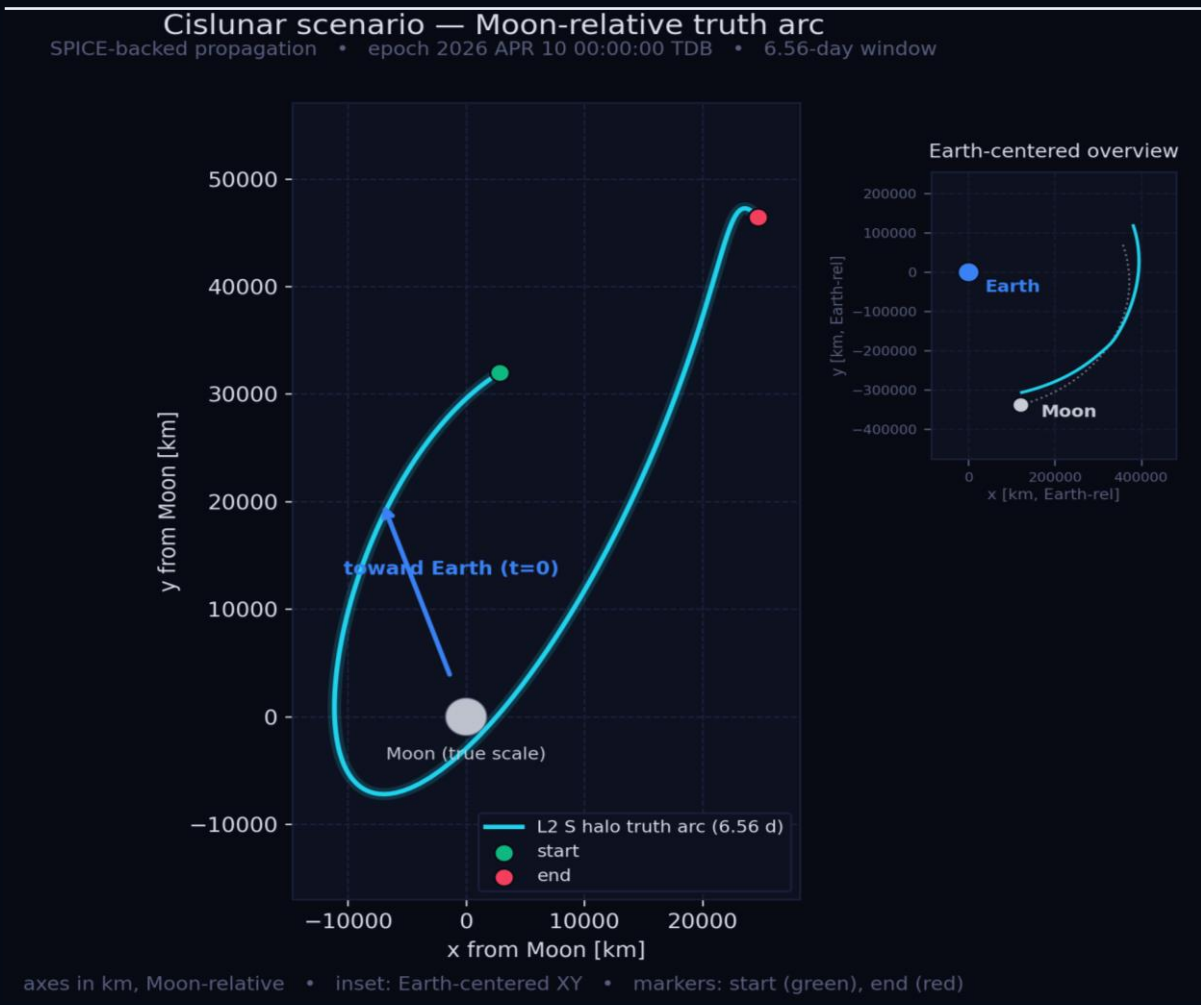
IEKF Measurement Update
Predict · Update (IEKF)

Without Phase 7
Fixed camera → target leaves FOV → measurements drop → estimator degrades

With Phase 7
Estimated LOS repoints camera → target stays in FOV → updates persist → stable filter

Cislunar Scenario: SPICE-Backed NRHO Propagation with Onboard LOS Geometry

The navigation pipeline is evaluated on a cislunar trajectory propagated from a periodic-orbit seed into a higher-fidelity Moon-relative scenario



Scenario

- Earth-Moon cislunar trajectory
- Moon-relative visualization
- SPICE-backed propagation used for higher-fidelity geometry

Propagation Model

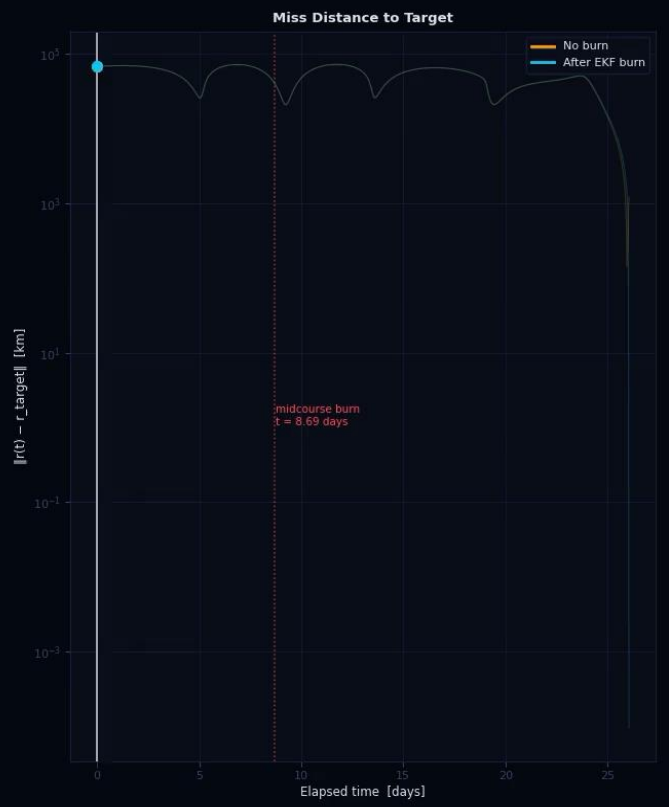
- State: spacecraft position and velocity
- Dynamics: truth trajectory propagated in a cislunar grav environment
- State transition info used later by estimator + targeting solve

Importance of this setup

- Gives realistic geometry for los sensing
- Defines where burns and corrections are applied
- Lets filter and guidance results be interpreted in a mission-like context

Can the Estimated State Produce a *Useful* Correction Burn?

A single burn is applied at correction time t_c; the key question is whether the burn computed from the estimated state meaningfully reduces terminal miss



Correction Problem

- Choose a correction burn at time t_c
- Use the available state estimate at t_c
- Reduce final miss relative to the desired target state at the end of the arc

Three Cases

- No correction
- Perfect-information burn
- IEKF-based burn

Reported Metrics

- Terminal miss
- Burn error magnitude
- Relative Δv inflation

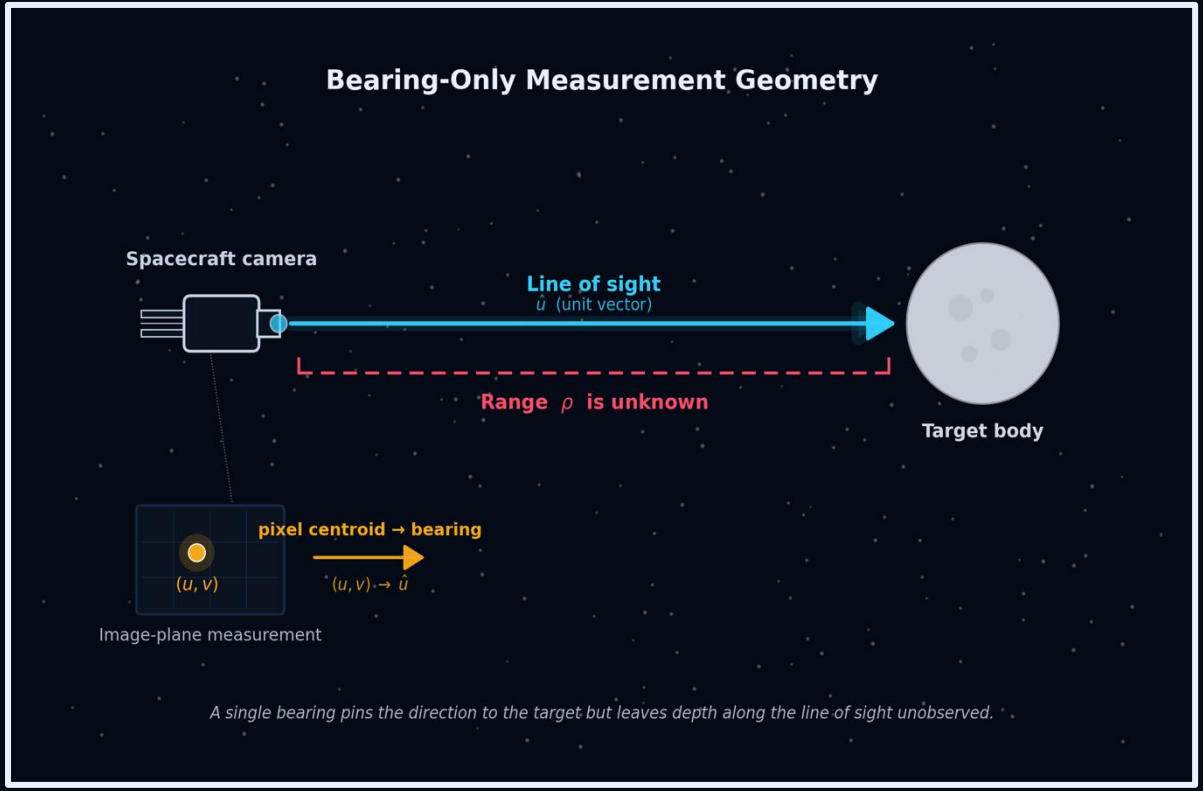
Definition Used Later:

$$Relative\ \Delta v\ inflation = \left(\frac{\| \Delta v_{IEKF} \|}{\| \Delta v_{perfect} \|} - 1 \right) \times 100\%$$

0% means the estimated-state burn matches the perfect-information burn in magnitude

A Camera Measures Line-of-Sight Direction, Not Range

The camera provides image-plane target location, which can be converted into a unit los vector, but the spacecraft-to-target distance is not directly observed



- **Measured directly:** image coordinates (u, v) in pixels
- **Converted measurement:** unit los vector $\boldsymbol{u}$
- **Not measured directly:** target range ρ

Bearing Measurement

$$\mathbf{u} = \left(\frac{\mathbf{r}_{body} - \mathbf{r}_{sc}}{\| \mathbf{r}_{body} - \mathbf{r}_{sc} \|} \right)$$

The observation constrains direction only

Why this is challenging

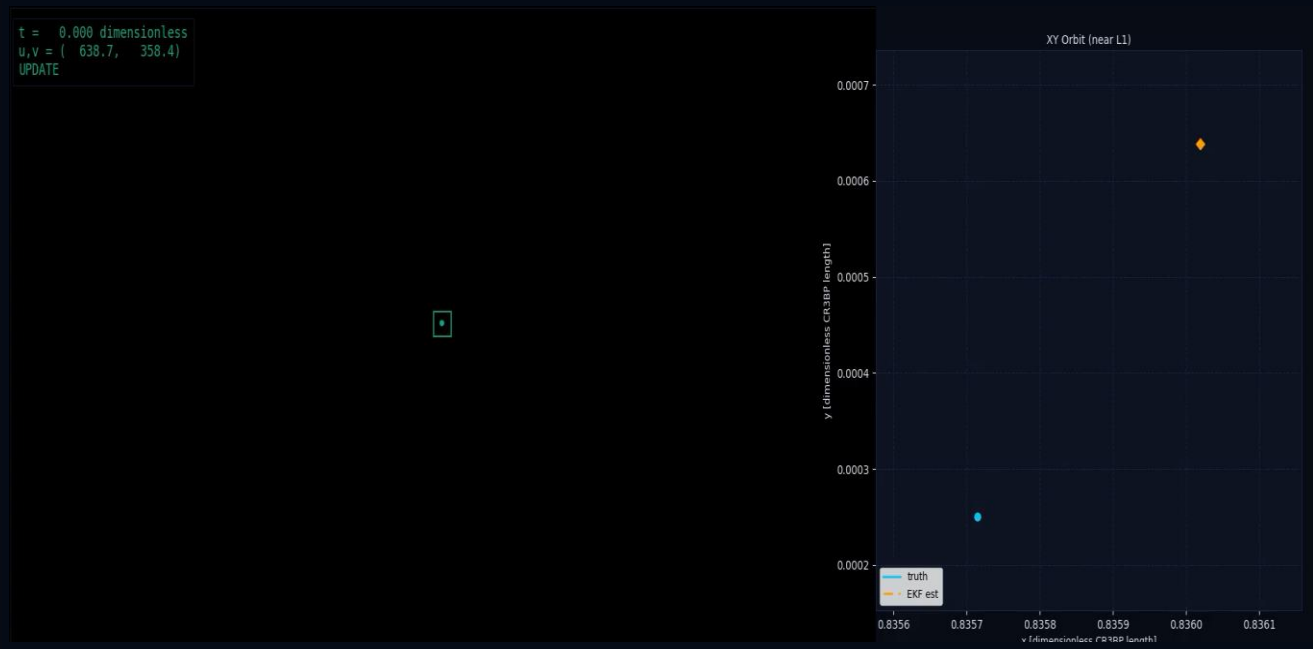
- Spacecraft state is 6-dimensional
- Bearing update is effectively 2-dimensional
- Distance must be recovered from dynamics + repeated geometry changes

Optical navigation is possible here because dynamics provide the missing structure that a single image does not

The Full Pixel-to-Bearing Pipeline Produces a Useable Navigation Estimate

The report summarizes estimation quality, while the demo shows the target remains measurable on the image plane during the arc

Phase 05 Demo

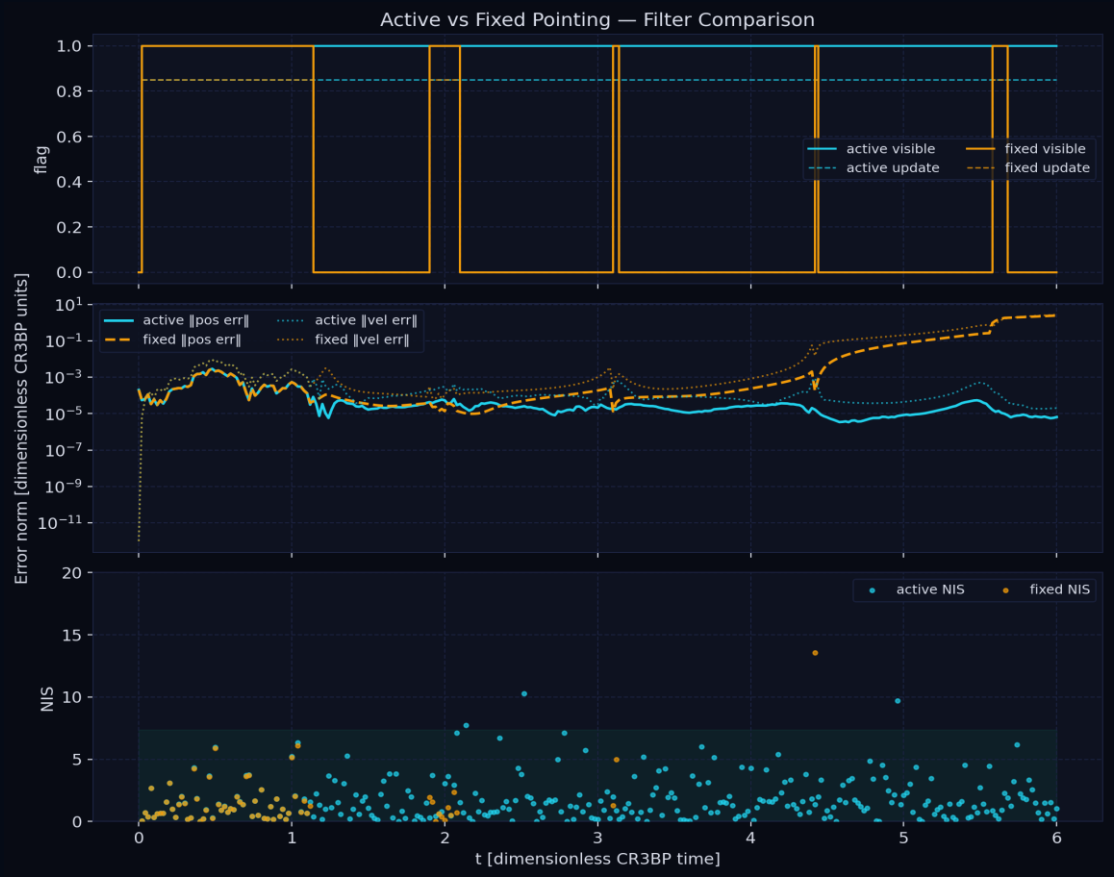


- Target remains measurable on the image plane
- Repeated pixel detections are converted to bearings
- Resulting estimate stays bounded and statistically interpretable

The pipeline depends on keeping the target measurable long enough for repeated pixel-to-bearing updates to improve the estimate

Keeping the Target in FOV Improves Measurement Availability

When the camera is repointed using the estimated LOS, the target remains visible longer, updates are sustained, and estimation quality improves relative to fixed pointing



What changes with active pointing?

- **Visibility increases:** target stays in FOV much longer!
- **Update rate increases:** more image measurements reach the IEKF
- **Error decreases:** active pointing maintains a much lower state-estimation error than fixed pointing

Fixed vs Active Results Summary

Visibility fraction: 0.246 → 0.997
Update fraction: 0.246 → 0.997
RMS position error: $5.44\times10^{-1} \rightarrow 4.12\times10^{-4}$
RMS velocity error: $5.84\times10^{-1} \rightarrow 1.38\times10^{-3}$

Active camera pointing becomes part of the navigation architecture because it preserves the measurements that the IEKF needs

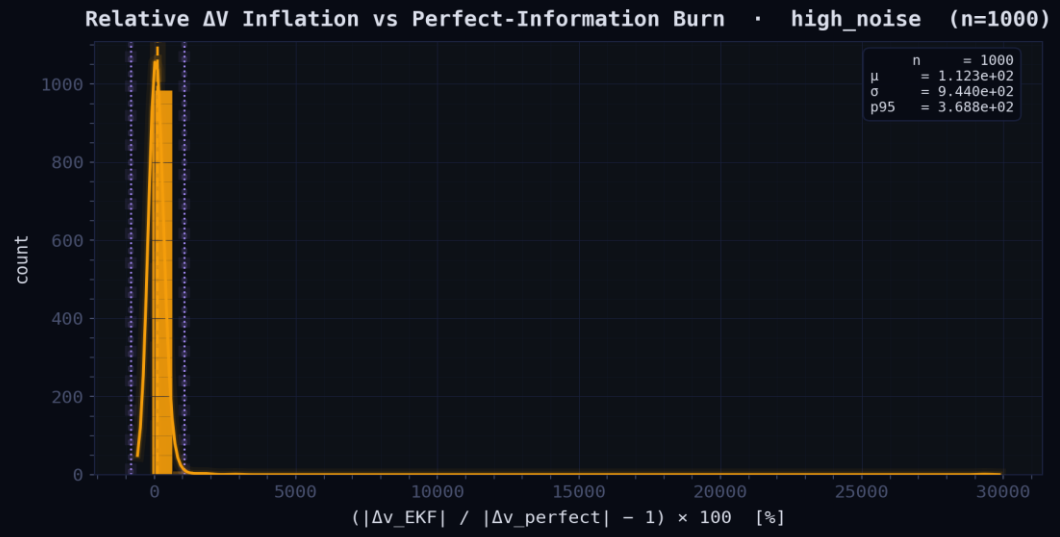
Across Many Trials, IEKF-Based Correction Stays Near the Perfect-Information Benchmark

MC trials quantify how much the estimated-state correction differs from the perfect-information solution in both burn magnitude and terminal performance

These Monte Carlo results are shown in dimensionless CR3BP units; histogram percentages are unitless.

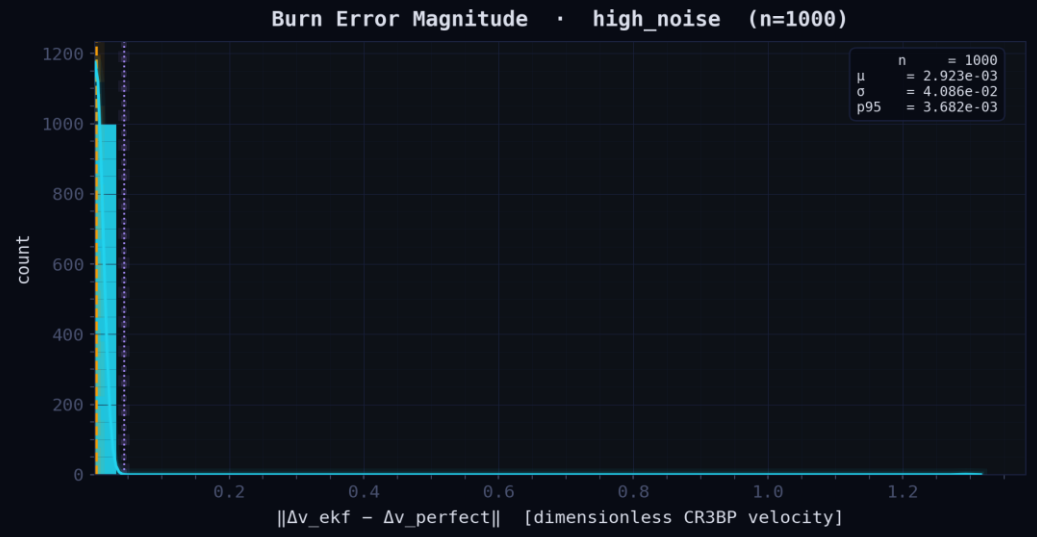
$$\text{Relative } \Delta v \text{ inflation} = \left(\frac{\| \Delta v_{IEKF} \|}{\| \Delta v_{perfect} \|} - 1\right) \times 100\%$$

A representative Monte Carlo trial shows the IEKF-based burn closely following the perfect-information correction geometry



The Δv inflation distribution is centered near zero, with a positive tail for harder trials.

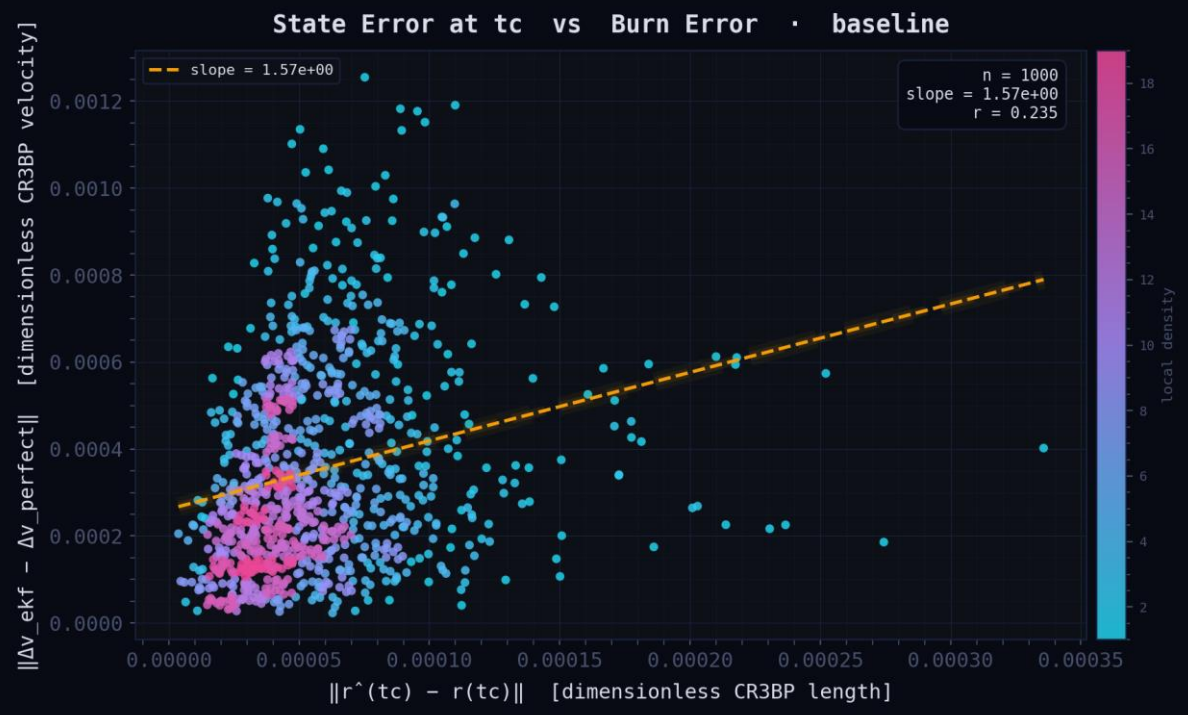
Absolute burn-error magnitude remains small relative to the correction scale across most trials.

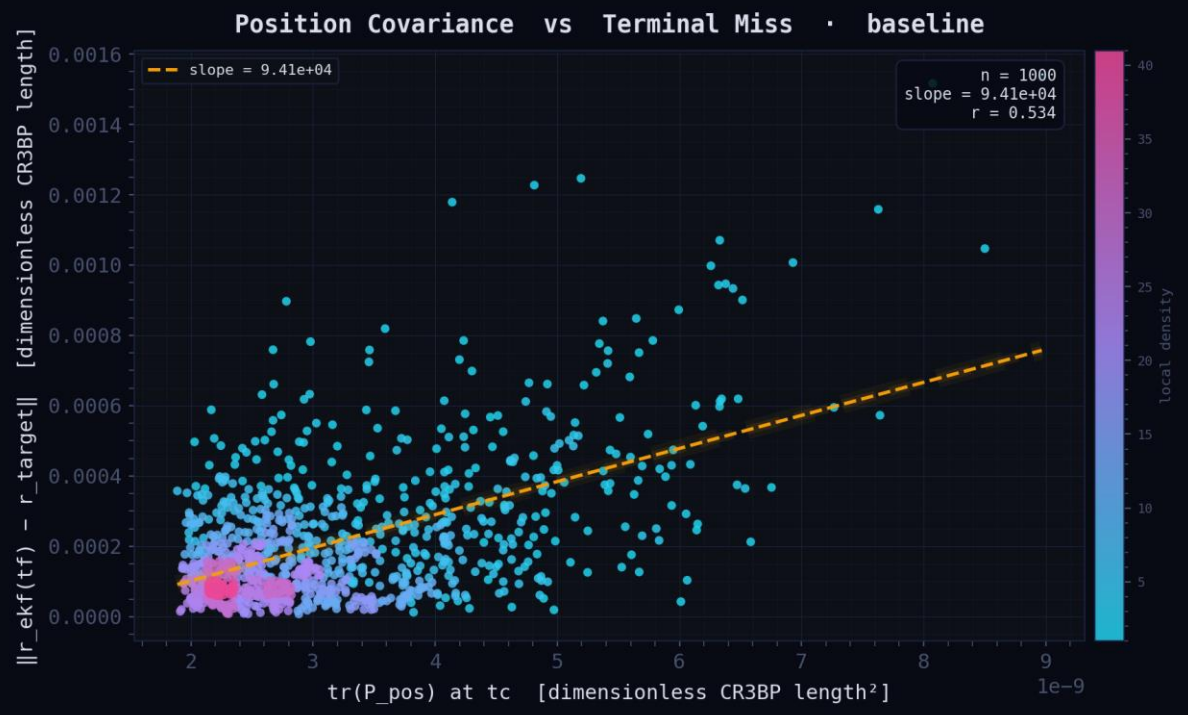


MC results show estimated-state correction usually remains close to perfect-info benchmark. Uncertainty creates a measurable spread in burn quality

Worse State Knowledge at Correction Time Leads to Worse Guidance Performance

Across MC trials, larger state uncertainty and larger estimation error at correction time tend to correlate with worse correction quality and larger terminal miss





Trials with larger position error at t_c tend to produce larger burn mismatch $\| \Delta v_{IEKF} \| - \| \Delta v_{perfect} \|$

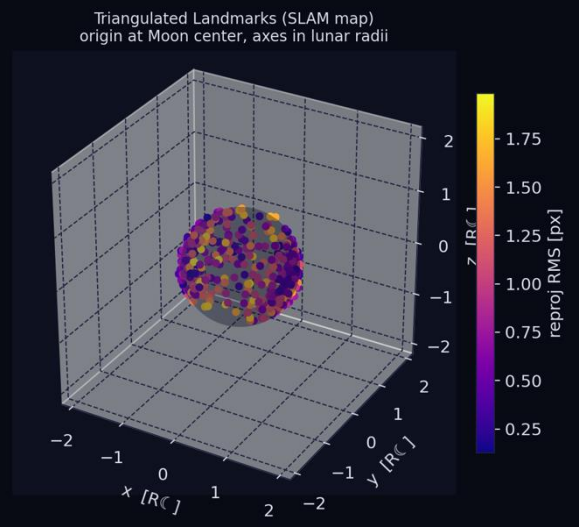
Larger covariance trace at t_c tends to correspond to larger terminal miss

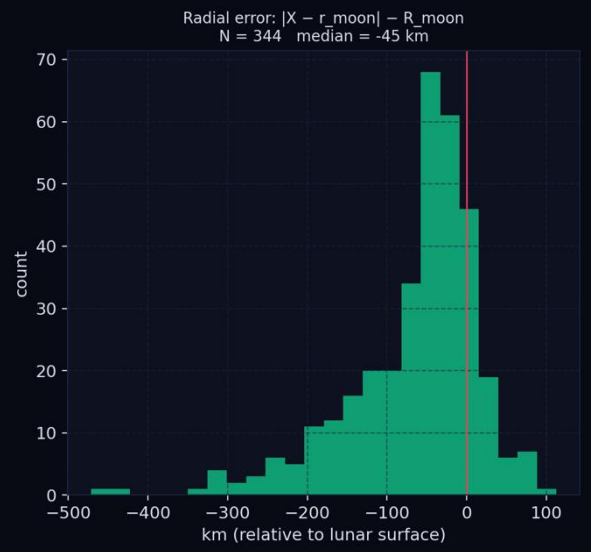
All quantities on this slide are in dimensionless CR3BP units except correlation statistics, which are unitless

The guidance outcome depends not just on whether the filter updates, but on how accurate and how certain the state estimate is at correction time

[Future work] Feature Tracking Extends the Vision Pipeline Beyond a Single Centroid

Instead of reducing each frame to one image-plane centroid, Phase 08 tracks multiple visual features and triangulates candidate landmarks, opening a path toward richer optical navigation





What Phase 08 Demonstrates

Demonstrates: multi-feature tracking and candidate landmark geometry

Does not yet demonstrate: a fully integrated landmark-aided nav filter

Validation + Literature Comparison

The current validated pipeline uses a monocular bearing-only measurement model, whereas much of the lunar optical-navigation literature uses richer landmark-based measurements. The validation question here is therefore whether bearing-only updates remain useful for correction in cislunar geometry, while Phase 08 points toward feature- and landmark-assisted navigation.

Reference	Scenario	Measurement type	Useful number/result	Relation to this work
Heidrich & Holzinger, 2025	Cislunar periodic orbit IOD	Angles-only / LOS	Robust cislunar angles-only orbit reconstruction over large observing gaps	Closest conceptual match to this cislunar, bearing-only framing
Federici et al., 2025	Low lunar orbit	Crater landmarks + EKF	~75 m transverse/normal, ~100 m radial, <20 cm/s at 3σ in benchmark case	Stronger measurement model than this current centroid-only pipeline
Pin-point lunar landing, 2014	Lunar landing	Feature/crater-based vision nav	Better than 100 m and 0.1 m/s	Richer landing-regime benchmark, not apples-to-apples
This project	Cislunar correction arc	Pixel centroid → bearing-only IEKF	Bounded estimation + useful correction + strong dependence on visibility/pointing	Weaker measurement model, stronger emphasis on measurement availability

This project does not yet target landmark-level absolute accuracy; instead, it validates that a weaker centroid/bearing-only pipeline can still produce useful correction decisions in cislunar geometry, and Phase 08 points toward the landmark-assisted regime studied in the literature

Conclusions: Bearing-Only Optical Navigation Can Support Useful Cislunar Correction

This project evaluated whether pixel-based optical measurements, processed through an IEKF and active-pointing pipeline, can provide state knowledge good enough to support midcourse correction in cislunar geometry.

(1) A camera is a viable nav sensor when direct support is weak

In cislunar space, image-based sensing can provide useful onboard nav information when direct GPS-like support is limited, but the measurements are fundamentally bearing-only

Using pixel-to-bearing conversion and iterated filter updates, the pipeline produced bounded and decision-useful state estimates under repeated measurement availability.

(2) Bearing-only updates can recover a usable state estimate

(3) Camera pointing policy is part of the navigation architecture

Active repointing greatly improved visibility, update fraction, and estimation quality relative to fixed pointing, showing that keeping the target in view is a first-order systems issue

Across representative and Monte Carlo trials, the IEKF-based correction stayed close to the perfect-information benchmark and materially reduced terminal miss

(4) The estimated state was good enough to support useful correction

The main result is not just that the filter runs. It is that a camera, an IEKF, and an active-pointing policy together form a usable cislunar optical-navigation pipeline

References and Resources

Core Texts

[1] Crassidis, J. L., & Junkins, J. L. Optimal Estimation of Dynamic Systems, 2nd ed., CRC Press, 2011

[2] Markley, F. L., & Crassidis, J. L. Fundamentals of Spacecraft Attitude Determination and Control, Springer, 2014

[3] Christian, J. A., & Robinson, S. B. Fundamentals of Spacecraft Optical Navigation, Wiley, 2022

[4] Hughes, P. C. Spacecraft Attitude Dynamics, Dover, 2004

Course / Academic Resources

UIUC AE 468: Optical Remote Sensing

UIUC CS 357: Numerical Methods

NASA Goddard: Optical Navigation Techniques (public materials)

Methods & Theory

[5] Maybeck, P. S. Stochastic Models, Estimation, and Control, Vol. 1, Academic Press

[6] Bar-Shalom, Y., Li, X. R., & Kirubarajan, T. Estimation with Applications to Tracking and Navigation, Wiley

Project Code

GitHub Repository: https://github.com/varshakrishnakumar/cislunar-optical-nav